Selected engineering work

Five projects, one method: **state the claim, name the evidence, and say where the evidence stops.** Every project below is public and every number in it is reproducible from a committed script.

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VOLLEY

primary project · systems engineering

An electromagnetic CubeSat deployer, analysed end to end and published with its own failures. **Sixty-five run sheets, each with its acceptance bands committed *before* the script that produces the number exists.** A magnetostatic 2-D FEM written from the weak form (scikit-fem, 141 k elements) and an independent 3-D solve (getdp, 274 105 DoF) check the thrust constant to 0.03 % and 0.059 %; CalculiX takes the structure, OpenFOAM the external aero at 581 779 cells, ngspice the pulse chain, GMAT R2022a the astrodynamics. **Two of the project's own competitive claims were withdrawn by its own analyses**, and the defect register carries 132 numbered entries with 53 still open.

Python, parametric CAD, FEM, CFD, circuit simulation, astrodynamics, control, configuration management
github.com/aaaaaaaaaaaavm/VOLLEY

BOLLEY

sibling study · the opposite premise

VOLLEY refuses to touch the satellite. **BOLLEY tests what that refusal costs** by letting a compatible CubeSat carry a few hundred grams of *passive* interface — no power, no electronics, nothing to command — so the launcher can delete the sled, the brake and the return stroke. Developed to the same standard, and it records the branches that failed: a buried return, a three-fin comb, a shared-pole quad-comb, and a first-generation design that passed a thin-sheet circuit model and then lost to a 66.7:1 copper-window deficit when exact CAD was drawn. Of 2 856 declared candidates, 77 clear every hard band.

Design-of-experiments over a declared candidate space, nonlinear magnetostatics, thermal and duty-cycle screening
github.com/aaaaaaaaaaaavm/BOLLEY

Orbital Deployment Trade Study

installable tool

A preliminary calculator for tangential deployment impulses, phase drift, host recoil and the rigid-body disturbance from a closed internal mass move — separated out because VOLLEY's astrodynamics record was mixing four questions that do not carry the same evidence. **It carries the boundary that embarrassed its own parent**: phase drift is not the cheapest route to phase, because release timing reaches 30° in 468 seconds at zero Δv while a commanded differential takes 1.38 days and then never stops drifting.

Two-body and rigid-body mechanics, packaging, CI | *does not do conjunction assessment or establish flight safety*
github.com/aaaaaaaaaaaavm/orbital-deployment-trade-study

Pulsed Linear Motor Design Lab

installable tool

A screening calculator for pulsed linear-motor stages: given moving mass, stroke, thrust constant, sheet current, force limit, source voltage and ESR, what constant-force velocity is reachable and can the source deliver the peak power? Reproduces the detailed VOLLEY reference to **0.005 m/s**, inside a declared 0.01 m/s regression band. **Its most useful output is a failure**: the source check closes at the 12 $\mathrm{m}\Omega$ modelled assumption and *fails* at the 116 $\mathrm{m}\Omega$ of a representative purchasable capacitor string.

Numerical modelling, regression banding, source-impedance analysis | *model-to-model agreement, not a measurement*
github.com/aaaaaaaaaaaavm/pulsed-linear-motor-design-lab

Engineering Evidence Toolkit

installable tool

A dependency-free command-line check for whether computational results, Markdown links and declared artifacts still agree with their recorded sources: `source files → numerical JSON → report links → declared artifact hashes`. Built after a long numerical project accumulated the ordinary failures — a PDF older than its source, copied JSON no script regenerates, figures from a superseded operating point. A declared pattern matching nothing is itself a failure, so a misspelt pattern cannot pass by checking nothing.

Consistency checking, not physics validation. A green result says the declared chain still holds; it says nothing about whether the first equation in it was right.
github.com/aaaaaaaaaaaavm/engineering-evidence-toolkit

What I would want a reader to check first. Not the headline numbers — the places where the work argues against itself. VOLLEY's register, and the three claims its own manuscript withdraws, are a better sample of how I work than any result on the front of it. Nothing in any of these projects has been built, fired, measured or flown, and every repository says so on its own front page rather than in a footnote.

All of it, and the rest
github.com/aaaaaaaaaaaavm